

Deployment Diagram

These diagrams are where the softwares meet hardware.

They are the physical floor plans of the data center.

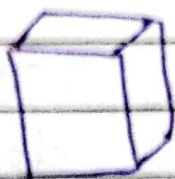
A deployment diagram maps the physical architecture of a system.

* It shows which physical devices run which physical files (Artifacts).

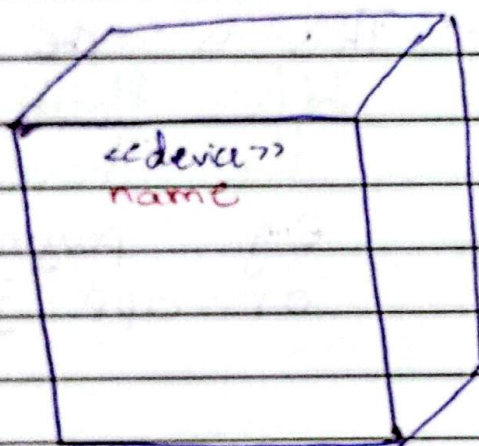
Structure:

1) Node:

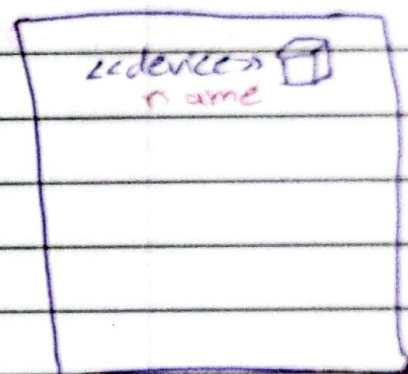
This is basically a 3D box



This is also called a **device node**.



OR



- both representation are correct.

NAME: This shows the device we are using e.g., web server, Database Server, personal Computer, Mobile Phone, iPhone 15, Application Server.

2) Artifact:

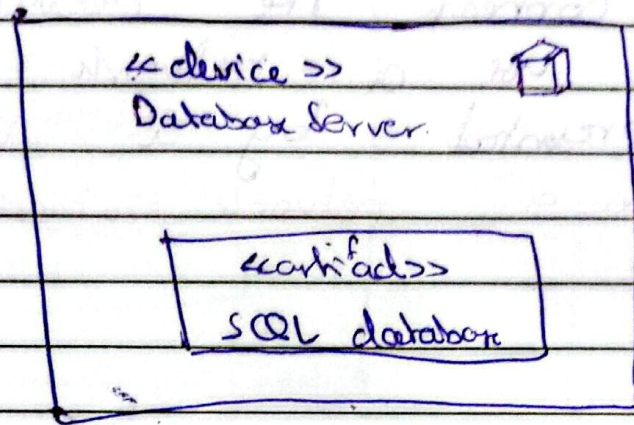
Remember from component diagram, there were components & artifact.

Component: This is the logical design of software blocks e.g. "payment processor", it represents idea & code of payment system.

Artifact: This is the actual file that sits on the hard drive. e.g. payment-service.dll, or app.jar.

These artifacts are added as boxes to their respective hardware node.

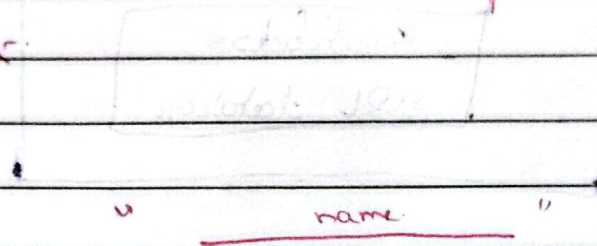
Example,



- artifacts exist on hardware so we only include them, NOT the components.

3) Communication Path:

To connect the device nodes we use a network path represented by a straight line with a network protocol written on it.



network protocols:

- when an app talks to server it can be

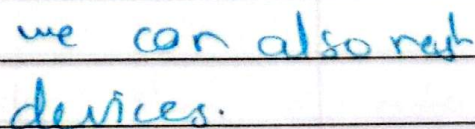
- HTTPS
- Port 443
- Port 80 ~~HTTP~~ (unencrypted HTTP)

- When any node talks to any DB node, we use ~~IP address~~ ~~because~~ ~~we can't~~ ~~clearing~~ ~~IP~~ ~~address~~ specific ports.

Like Port 3306 is for
«mysql protocol».

- you can also just write « oracle protocols »
« $\mathcal{B}$ @ Server Protocol ».

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Important:-

- An "execution environment", ^{node} can also be used in a deployment diagram.
- This shows the software container that hosts your files.
e.g windows OS, Docker container, JVM

